ESW index-free $\longrightarrow$ Weinberg–Srednicki two-component

Dictionary for the $\mathcal{N}=4$ Costello–BV program (Step 2, directive R1)

2026-10-28

1 Sources and base entries

item	value	evidence
spinor metric	$\varepsilon^{12}=\varepsilon^{\dot1\dot2}=+1$, $\varepsilon_{12}=\varepsilon_{\dot1\dot2}=-1$	S-SPN-2: Srednicki (35.1)
raising, lowering	$\psi^\alpha=\varepsilon^{\alpha\beta}\psi_\beta$, $\psi_\alpha=\varepsilon_{\alpha\beta}\psi^\beta$, same for dotted	S-SPN-3: (34.24), (34.25)
contractions	$\chi\psi=\chi^\alpha\psi_\alpha$, $\bar\chi\bar\psi=\bar\chi_{\dot\alpha}\bar\psi^{\dot\alpha}$	S-SPN-4: (35.23)
σ, $\bar\sigma$	$\sigma^\mu_{\alpha\dot\alpha}=(1,\vec\sigma)$, $\bar\sigma^{\mu\dot\alpha\alpha}=\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^\mu_{\beta\dot\beta}=(1,-\vec\sigma)$	S-SPN-5: (34.28), (35.2), (35.3)
metric, Clifford	$g=\mathrm{diag}(-1,1,1,1)$, $\sigma^\mu\bar\sigma^\nu+\sigma^\nu\bar\sigma^\mu=-2g^{\mu\nu}$	S-MET-1 (2.4), S-MET-5 (36.8)
Weyl block of γ	$\gamma^\mu_S=\begin{pmatrix}0&\sigma^\mu\\ \bar\sigma^\mu&0\end{pmatrix}$ on $(\psi_\alpha,\bar\chi^{\dot\alpha})$, $\gamma_{5S}=\mathrm{diag}(-1,-1,1,1)$	S-SPN-7 (36.44), S-SPN-8 (36.7)
Weinberg γ	$\gamma^\mu_W=-i\gamma^\mu_S$ (same Weyl basis), $\gamma_{5W}=-\gamma_{5S}$	weinberg-qtf-1995 D1, D2
gauge (Srednicki)	$D_\mu=\partial_\mu-igA^a_\mu T^a$, $F^c=\partial A^c-\partial A^c+gf^{abc}A^aA^b$	S-GRP-5 (69.12), S-GRP-6 (69.22)
gauge (Weinberg)	$t_\alpha=gT^a$, $C_{\alpha\beta\gamma}=gf^{abc}$, $\Lambda_W=-\Gamma_S$	weinberg-qtf-1995 D9

The letters $\alpha,\dot\alpha$ of [SW, (17)] replace Srednicki's $a,\dot a$ (renaming). The Euclidean entries SIG, REP, CONF of STANDARD.md are not used; the Euclidean adaptation below is derived from the realisation of engine B (`engine_b/realization.py`) and checked against the realisation of engine A (`engine_a/cliff.py` = `classical/n4_model.py`). ESW-side formulas are quoted from the manuscript with their labels.

2 Euclidean two-component realisation

Definition 2.1 (Wick map and Euclidean σ)**.** For $v\in V$ with Euclidean components v^m, $m=1,2,3,4$, the Lorentzian vector is $v^0=-iv^4$, $v^k=v^k$ $(k=1,2,3)$; in particular $x^4=it$ and $\partial_0=i\partial_4$. Define

$$\sigma^m_{E\,\alpha\dot\alpha}=(\sigma^1,\sigma^2,\sigma^3,\,i\,1),\qquad \bar\sigma^{m\dot\alpha\alpha}_E=(-\sigma^1,-\sigma^2,-\sigma^3,\,i\,1).$$

Then

$$\sigma^\mu v_\mu=\sigma^m_E v_m,\qquad \bar\sigma^\mu v_\mu=\bar\sigma^m_E v_m,\qquad \sigma^m_E\bar\sigma^n_E+\sigma^n_E\bar\sigma^m_E=-2\delta^{mn},\qquad \bar\sigma^m_E\sigma^n_E+\bar\sigma^n_E\sigma^m_E=-2\delta^{mn},$$

$$\bar\sigma^{m\dot\alpha\alpha}_E=\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^m_{E\,\beta\dot\beta},\qquad \bar\sigma^m_E=-(\sigma^m_E)^\dagger.$$

Below $\sigma^m\equiv\sigma^m_E$, $\bar\sigma^m\equiv\bar\sigma^m_E$, Euclidean indices up or down alike. Orientation of V: $dx^1\wedge dx^2\wedge dx^3\wedge dx^4$.

Definition 2.2 ($SU(4)$ blocks)**.** $W\ni w^i$ (upper index), $W^*\ni w_i$ (lower index), $i=1,\dots,4$. On $W^\perp_{\mathbb{R}}$ with orthonormal basis e_I, $I=1,\dots,6$:

$$\Sigma^I_{ij}=(\eta^1,\eta^2,\eta^3,i\bar\eta^1,i\bar\eta^2,i\bar\eta^3)_{ij},\qquad \bar\Sigma^I=(\Sigma^I)^\dagger=(-\eta^1,-\eta^2,-\eta^3,i\bar\eta^1,i\bar\eta^2,i\bar\eta^3),$$

$\eta^a_{bc} = \bar\eta^a_{bc} = \epsilon_{abc}$ $(b,c \le 3)$, $\eta^a_{b4} = -\eta^a_{4b} = \delta_{ab}$, $\bar\eta^a_{b4} = -\bar\eta^a_{4b} = -\delta_{ab}$ (engine B, `thooft`). With $\varepsilon_{1234} = \varepsilon^{1234} = 1$:

$$\Sigma^I_{ij} = -\Sigma^I_{ji}, \qquad \tfrac12 \varepsilon^{ijkl}\Sigma^I_{kl} = -\bar\Sigma^{I\,ij}, \qquad \bar\Sigma^I\Sigma^J + \bar\Sigma^J\Sigma^I = 2\delta^{IJ}, \qquad \operatorname{tr}(\bar\Sigma^I\Sigma^J) = 4\delta^{IJ},$$

$$\sum_I \Sigma^I_{ij}\Sigma^I_{kl} = 2\varepsilon_{ijkl}, \qquad \sum_I \bar\Sigma^{I\,ij}\bar\Sigma^{I\,kl} = 2\varepsilon^{ijkl}, \qquad \sum_I \bar\Sigma^{I\,ij}\Sigma^I_{kl} = -2\big(\delta^i_k\delta^j_l - \delta^i_l\delta^j_k\big).$$

For $u \in W^\perp_{\mathbb R}$ (components u_I): $u_{ij} = u_I\Sigma^I_{ij}$, $u^{ij} = -u_I\bar\Sigma^{I\,ij} = \frac12\varepsilon^{ijkl}u_{kl}$, and

$$(u,w) = \tfrac14 u^{ij}w_{ij}, \qquad u_I = \tfrac14 \bar\Sigma^{I\,ji}u_{ij}, \qquad u^{ik}w_{kj} + w^{ik}u_{kj} = -2(u,w)\,\delta^i_j.$$

Theorem 2.3 (Two-component form of Σ, Σ^*, ρ, $(-,-)$, Γ)**.** Write $\xi \in \Sigma = S_+ \otimes W \oplus S_- \otimes W^*$ as $(\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$ and $\eta \in \Sigma^* = S_+ \otimes W^* \oplus S_- \otimes W$ as $(\eta^\alpha{}_i, \bar\eta^{\dot\alpha i})$; S_+ carries undotted, S_- dotted indices. For $v \in V$ and $w \in W^\perp_{\mathbb R}$:

$$(\rho(v)\xi)^\alpha{}_i = i\,v_m\,\bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}, \qquad (\rho(v)\xi)^{\dot\alpha i} = -i\,v_m\,\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i, \tag{R1}$$

$$(\rho(v)\eta)_\alpha{}^i = -i\,v_m\,\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad (\rho(v)\eta)_{\dot\alpha i} = i\,v_m\,\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}, \tag{R2}$$

$$(\rho(w)\xi)^\alpha{}_j = \varepsilon^{\alpha\beta}w_{ji}\,\xi_\beta{}^i = w_{ji}\,\xi^{\alpha i}, \qquad (\rho(w)\xi)^{\dot\alpha j} = \varepsilon^{\dot\alpha\dot\beta}w^{ji}\,\bar\xi_{\dot\beta i} = w^{ji}\,\bar\xi^{\dot\alpha}{}_i, \tag{R3}$$

$$(\rho(w)\eta)_\alpha{}^k = -w^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j = -w^{kj}\eta_{\alpha j}, \qquad (\rho(w)\eta)_{\dot\beta k} = -w_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j} = -w_{kj}\bar\eta_{\dot\beta}{}^j, \tag{R4}$$

$$(\xi,\eta) = \xi_\alpha{}^i\eta^\alpha{}_i - \bar\xi_{\dot\alpha i}\bar\eta^{\dot\alpha i} = -(\xi^i\eta_i) - (\bar\xi_i\bar\eta^i), \tag{P1}$$

$$\Gamma(\xi_1,\xi_2)\big|^m_V = i\big(\bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i + \bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i\big), \tag{G1}$$

$$\begin{aligned}\Gamma(\xi_1,\xi_2)_{kl} &= 2\varepsilon_{jikl}\,\xi_{1\alpha}{}^j\varepsilon^{\alpha\beta}\xi_{2\beta}{}^i - 2\big(\bar\xi_{1\dot\alpha k}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta l} - \bar\xi_{1\dot\alpha l}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta k}\big)\\ &= -2\varepsilon_{jikl}(\xi_1{}^j\xi_2{}^i) - 2(\bar\xi_{1k}\bar\xi_{2l}) + 2(\bar\xi_{1l}\bar\xi_{2k}),\end{aligned} \tag{G2}$$

$$\Gamma(\eta_1,\eta_2)\big|^m_V = -i\big(\eta_2{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_1^{\dot\alpha i} + \eta_1{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_2^{\dot\alpha i}\big), \tag{G3}$$

$$\Gamma(\eta_1,\eta_2)^{kl} = -2\varepsilon^{klpj}\varepsilon_{\alpha\beta}\eta_1{}^\beta{}_j\eta_2{}^\alpha{}_p + 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha l}\bar\eta_2^{\dot\beta k} - \varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha k}\bar\eta_2^{\dot\beta l}\big), \tag{G4}$$

where $\Gamma_{kl} = \Gamma^I\Sigma^I_{kl}$, $\Gamma^{kl} = -\Gamma^I\bar\Sigma^{I\,kl}$, $(\Gamma,u) = \Gamma^m u_m + \frac14\Gamma_{ij}u^{ij}$, Γ defined by $(\rho(v)Q_1,Q_2) = (v,\Gamma(Q_1,Q_2))$ [ESW, (28)], ch01 Def c1-spaces. The data satisfy:

- $\rho(v_1)\rho(v_2) + \rho(v_2)\rho(v_1) = 2(v_1,v_2)$ on Σ and Σ^*, $(\rho(v)Q_1,Q_2) = (Q_1,\rho(v)Q_2)$ (ch01 Def c1-spaces);
- Γ symmetric; $\rho(\Gamma(Q_1,Q_2))Q_3$+cyclic $= 0$ [ESW, Assumption 3.1]; $\Gamma(Q_1,\rho(X)Q_2)+\Gamma(Q_2,\rho(X)Q_1) = -2\iota_{\Gamma(Q_1,Q_2)}X$ [ESW, Prop 2.1];
- $\rho(e_1e_2e_3e_4) = +1$ on $S_+\otimes W$, -1 on $S_-\otimes W^*$ (ch01 Prop c1-chirality); on the Dirac column $(\psi_\alpha, \bar\chi^{\dot\alpha})$, $\rho(v) = -i\gamma^\mu_S v_\mu = \gamma^\mu_W v_\mu$ and $\gamma^1_E\gamma^2_E\gamma^3_E\gamma^4_E = -\gamma_{5S}$: undotted $= S_+ =$ Srednicki left-handed $(2,1)$;
- $\xi_\alpha{}^i = u_\alpha w^i$, $\bar\xi = 0 \Rightarrow \Gamma(\xi,\xi) = 0$.

Proposition 2.4 (Identification with the engines)**.**

- Engine B: with the vector map $(m = 1,2,3,4;\ I) \mapsto (1,2,3,0;\ 3+I)$ and the basis change $\bar\xi_{\dot\alpha i} = \varepsilon_{\dot\alpha\dot\beta}\bar\xi^{\dot\beta}{}_i$ (column $\bar\xi^{\dot\beta}{}_i$ of engine B), $\eta^\alpha{}_i = \varepsilon^{\alpha\beta}\eta_{\beta i}$ (column $\eta_{\beta i}$ of engine B), (R1)–(R4) are engine B's ρ entry by entry, and (P1) is engine B's pairing with its normalisation (first nonzero entry $+1$), factor 1.
- Engine A = classical model: with the vector map $M \mapsto M$ ($m = M+1$ for $M \le 3$, $I = M-3$) there are $U\colon \Sigma \to \Sigma_A$, $U^*\colon \Sigma^* \to \Sigma^*_A$, unique up to a common sign, with $\rho_A(e_M)U = U^*\rho(e_M)$, $\rho_A(e_M)U^* = U\rho(e_M)$, $(Ua, U^*b)_A = (a,b)$; U maps $S_+\otimes W$ to the $+1$ eigenspace of $\rho_A(e_0e_1e_2e_3)$.

Remark 2.5 (Name clash, R3)**.** $\xi, \bar\xi$ (components of an element $Q_1 \in \Sigma$ of ch02) and $\eta, \bar\eta$ (components of $Q_2 \in \Sigma^*$) are spinor parameters. The one-loop components Q_1, S_1, D_1, K_1 of directive R3/R4 are unrelated objects.

3 Fields

ESW (ch01 Def c1-fields)	two-component	weight	parity
$A\|_V$	A_m, $m=1,\dots,4$	1	even
$\iota_w A$, $w\in W_{\mathbb{R}}^{\perp}$	A_I; $A_{ij}=A_I\Sigma^I_{ij}$, $A^{ij}=-A_I\bar\Sigma^{I\,ij}=\frac12\varepsilon^{ijkl}A_{kl}$; $\iota_wA=\frac14 w^{ij}A_{ij}$	1	even
$\lambda\in\Pi\Sigma\otimes\mathfrak{g}$	$\lambda_\alpha{}^i$ $(S_+\otimes W)$, $\bar\lambda_{\dot\alpha i}$ $(S_-\otimes W^*)$	$\frac32$	odd
c	c	0	odd
A^*	A^*_m; $A^*_{ij}=A^*_I\Sigma^I_{ij}$, $A^{*ij}=-A^*_I\bar\Sigma^{I\,ij}$	3	odd
$\lambda^*\in\Pi\Sigma^*\otimes\mathfrak{g}$	$\lambda^{*\alpha}{}_i$ $(S_+\otimes W^*)$, $\bar\lambda^{*\dot\alpha i}$ $(S_-\otimes W)$	$\frac52$	even
c^*	c^*	4	even
$Q_1\in\Sigma$	$\xi_\alpha{}^i$ ([SW,(17)] block $(3,1)$, Q^i_α), $\bar\xi_{\dot\alpha i}$ (block $(2,3)$, $\bar Q_{\dot\alpha i}$)		
$Q_2\in\Sigma^*$	$\eta^\alpha{}_i$ (block $(1,3)$, $S^\alpha{}_i$), $\bar\eta^{\dot\alpha i}$ (block $(3,2)$, $\bar S^{\dot\alpha i}$)		

$(-,-)$ on $\mathfrak{g}$ is kept; in every bilinear below it contracts the $\mathfrak{g}$ factors, in the order written. $\bar\lambda$ is an independent field (complexified Euclidean theory); on a Lorentzian real slice it is Srednicki's $\lambda^\dagger$ (S-CONJ-1). Real slice of the scalars (rem c1-real, A_I real): complex conjugate of $A_{ij}=A^{ij}$.

Proposition 3.1 (Pairings of fields)**.**

$$(A,A^*)=(A_m,A^*_m)+\tfrac14(A^{ij},A^*_{ij}),\qquad(\lambda,\lambda^*)=(\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i)-(\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i}),\qquad(c^*,c).$$

4 Action, Q, Q^{GF}, propagators

Theorem 4.1 (Action, ch01 eq:c1-action)**.** With $F_{mn}=\partial_mA_n-\partial_nA_m+[A_m,A_n]$, $D_m=\partial_m+[A_m,-]$,

$$\begin{aligned}S=\int d^4x\Big\{&-\tfrac18(F_{mn},F_{mn})-\tfrac{1}{16}(D_mA^{ij},D_mA_{ij})-\tfrac{1}{128}\big([A^{ij},A^{kl}],[A_{ij},A_{kl}]\big)\\&+\tfrac{i}{2}\big(\lambda_\alpha{}^i,\bar\sigma^{m\dot\alpha\alpha}D_m\bar\lambda_{\dot\alpha i}\big)+\tfrac{i}{2}\big(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}D_m\lambda_\alpha{}^i\big)+\tfrac12\big(\lambda_\alpha{}^j,\varepsilon^{\alpha\beta}[A_{ji},\lambda_\beta{}^i]\big)-\tfrac12\big(\bar\lambda_{\dot\alpha j},\varepsilon^{\dot\alpha\dot\beta}[A^{ji},\bar\lambda_{\dot\beta i}]\big)\\&+(D_mc,A^*_m)+\tfrac14([A^{ij},c],A^*_{ij})+([c,\lambda_\alpha{}^i],\lambda^{*\alpha}{}_i)-([c,\bar\lambda_{\dot\alpha i}],\bar\lambda^{*\dot\alpha i})+\tfrac12([c,c],c^*)\Big\},\end{aligned}$$

pointwise equal to the density of ch01 eq:c1-action. Integrated, with Srednicki contractions,

$$\begin{aligned}\int\tfrac12(\lambda,\not{d}_A\lambda)&=\int\Big\{i\big(\bar\lambda_i\bar\sigma^mD_m\lambda^i\big)-\tfrac12\big(\lambda^j[A_{ji},\lambda^i]\big)-\tfrac12\big(\bar\lambda_j[A^{ji},\bar\lambda_i]\big)\Big\},\\\int(\lambda,\not{d}\lambda)&=2i\int(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i),\\(F,F)&=\tfrac12(F_{mn},F_{mn})+\tfrac14(D_mA^{ij},D_mA_{ij})+\tfrac{1}{32}\big([A^{ij},A^{kl}],[A_{ij},A_{kl}]\big),\\\tfrac12\sum_{I,J}([A_I,A_J],[A_I,A_J])&=\tfrac{1}{32}\big([A^{ij},A^{kl}],[A_{ij},A_{kl}]\big).\end{aligned}$$

Corollary 4.2 (Wick image of Srednicki's Lagrangian)**.** For $\mathfrak{g}=\mathfrak{su}(N)$ with $(X,Y)=-\frac{4\hbar}{g^2}\mathrm{tr}(XY)$ (ch06 rem c6-konishi) and $A_m=-igA^a_mT^a$, $A_I=-igA^a_IT^a$, $\lambda=-\frac{ig}{\sqrt2}\lambda^aT^a$ ($\mathrm{tr}(T^aT^b)=\frac12\delta^{ab}$, S-GRP-2 (69.8)): $(X,Y)=2\hbar X^aY^a$, $F_{mn}=-igF^a_{mn}T^a$ with F^a of S-GRP-6, D_m the adjoint form of S-GRP-5, and

$$\frac{S}{\hbar}\supset\int d^4x\Big\{-\tfrac14F^a_{mn}F^a_{mn}-\tfrac12(D_mA_I)^a(D_mA_I)^a+i\bar\lambda^a\bar\sigma^mD_m\lambda^a\Big\}=\int d^4x\;\mathcal{L}_{\mathrm{Sred}}\big|_{x^0=-ix^4},$$

$$e^{iS_L}=e^{S/\hbar},$$

$\mathcal{L}_{\mathrm{Sred}}=-\frac14F^{a\mu\nu}F^a_{\mu\nu}-\frac12D^\mu A_ID_\mu A_I+i\lambda^{\dagger a}\bar\sigma^\mu D_\mu\lambda^a$ (S-ACT-9 (69.16), S-ACT-7 (3.9), S-ACT-12 (95.77)), using $F_{\mu\nu}F^{\mu\nu}=F_{mn}F_{mn}$ and $\bar\sigma^\mu\partial_\mu=\bar\sigma^m_E\partial_m$.

Proposition 4.3 (Q and Q^{GF}, ch01 Prop c1-Q, ch03 Def c3-qgf)**.**

slot	Q	lands in
c	$-\partial_m c$	A_m
A_m	$\frac{1}{2}(\partial^2 A_m - \partial_m \partial_n A_n)$	A^*_m
A_{ij}	$\frac{1}{2}\partial^2 A_{ij}$	A^*_{ij}
A^*_m	$-\partial_m A^*_m$	c^*
$\lambda_\alpha{}^i$	$-i\bar\sigma^{m\dot\alpha\alpha}\partial_m \lambda_\alpha{}^i$	$\bar\lambda^{*\dot\alpha i}$
$\bar\lambda_{\dot\alpha i}$	$i\,\partial_m \bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}$	$\lambda^{*\alpha}{}_i$

slot	Q^{GF}	lands in
$A^*_m,\ A^*_{ij}$	$-2A^*_m,\ -2A^*_{ij}$	$A_m,\ A_{ij}$
A_m	$\partial_m A_m$	c
c^*	$\partial_m c^*$	A^*_m
$\lambda^{*\alpha}{}_i$	$-i\,\partial_m \lambda^{*\alpha}{}_i \sigma^m_{\alpha\dot\alpha}$	$\bar\lambda_{\dot\alpha i}$
$\bar\lambda^{*\dot\alpha i}$	$i\,\sigma^m_{\alpha\dot\alpha}\partial_m \bar\lambda^{*\dot\alpha i}$	$\lambda_\alpha{}^i$

$H = [Q, Q^{GF}] = -\partial^2$ on every component ($|p|^2$ in momentum space).

Proposition 4.4 (Propagator $P(\varepsilon,\infty)$, ch03 Prop c3-P)**.** In momentum space each component is the fibre factor times $e^{-\varepsilon|p|^2}/|p|^2$:

$$\langle A_m A_n\rangle :\ 2(-,-)^{-1}\delta_{mn}, \qquad \langle A_{ij}A_{kl}\rangle :\ 4(-,-)^{-1}\varepsilon_{ijkl}, \qquad \langle A^{ij}A^{kl}\rangle :\ 4(-,-)^{-1}\varepsilon^{ijkl},$$
$$\langle A_{ij}A^{kl}\rangle :\ 4(-,-)^{-1}(\delta^k_i\delta^l_j - \delta^l_i\delta^k_j), \qquad \langle c\,A^*_m\rangle = \langle A^*_m\,c\rangle :\ -(-,-)^{-1}\,ip_m,$$
$$\langle \lambda_\alpha{}^i\,\bar\lambda_{\dot\beta j}\rangle :\ -(-,-)^{-1}p_m\sigma^m_{\alpha\dot\beta}\delta^i_j, \qquad \langle \bar\lambda_{\dot\alpha i}\,\lambda_\beta{}^j\rangle :\ -(-,-)^{-1}p_m\sigma^m_{\beta\dot\alpha}\delta^j_i,$$

$\langle\lambda\lambda\rangle = \langle\bar\lambda\bar\lambda\rangle = 0$; the $\lambda\otimes\lambda$ entries are $i\rho(p)|_{\Sigma^*\to\Sigma}$ read in $\Sigma\otimes\Sigma$ through (P1).

5 $\mathfrak{psl}(4|4)$

Definition 5.1 (Generators and Killing spinors, ch02 Def c2-even, c2-odd)**.**

generator	$v^m(x)$	$X_{mn} = \partial_m v_n - \partial_n v_m$	$\operatorname{div} v$	$r^i{}_j$
translation b ($P_m = \partial_m$)	b^m	0	0	0
rotation $\omega_{mn} = -\omega_{nm}$	$\omega_{mn}x_n$	$-2\omega_{mn}$	0	0
$\Delta = 2E$	$2x^m$	0	8	0
special conformal b	$x^2 b^m - 2(b\cdot x)x^m$	$4(x_m b_n - x_n b_m)$	$-8\,b\cdot x$	0
$R\in\mathfrak{so}(W^\perp_{\mathbb{R}})$, $v(y)^I = R_{IJ}y_J$	0	0	0	$\frac{1}{4}R_{IJ}(\bar\Sigma^J\Sigma^I)^i{}_j$

$r\in\mathfrak{sl}(4)$ is traceless. Odd part: $\xi\in\Sigma$ realised by the constant spinor, $\eta\in\Sigma^*$ by $\rho(x)\eta$; the conformal Killing spinor $x\mapsto \xi+\rho(x)\eta$ is

$$\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad \bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}. \tag{K1}$$

Theorem 5.2 (Brackets, ch02 eq:c2-evenodd, eq:c2-oddodd, Prop c2-special)**.**

$$[\epsilon_1,\epsilon_2] = \Big(x\mapsto i\big(\bar\epsilon_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{2\alpha}{}^i + \bar\epsilon_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{1\alpha}{}^i\big)\partial_m\Big) + r, \tag{B1}$$

$$r^k{}_i = 2\big(\xi_{1\alpha}{}^k\eta_2{}^\alpha{}_i + \bar\xi_{1\dot\alpha i}\bar\eta_2^{\dot\alpha k}\big) + 2\big(\xi_{2\alpha}{}^k\eta_1{}^\alpha{}_i + \bar\xi_{2\dot\alpha i}\bar\eta_1^{\dot\alpha k}\big) - \text{trace}, \tag{B2}$$

for $\epsilon_a = \xi_a + \rho(x)\eta_a$; the V-part of $[\xi,\rho(x)\eta]$ is $\frac{1}{2}(\xi,\eta)\Delta$ plus a rotation, (ξ,η) as in (P1). For an even generator,

$$\begin{aligned}[v,\epsilon]_\alpha{}^i &= v^n\partial_n\epsilon_\alpha{}^i - \tfrac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\epsilon_\beta{}^i - \tfrac{1}{8}\operatorname{div} v\,\epsilon_\alpha{}^i + r^i{}_j\epsilon_\alpha{}^j,\\ [v,\epsilon]_{\dot\alpha i} &= v^n\partial_n\bar\epsilon_{\dot\alpha i} + \tfrac{1}{8}X_{mn}\bar\epsilon_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} - \tfrac{1}{8}\operatorname{div} v\,\bar\epsilon_{\dot\alpha i} - \bar\epsilon_{\dot\alpha j}r^j{}_i.\end{aligned} \tag{B3}$$

Special cases: $[\Delta,\xi] = -\xi$, $[\Delta,\eta] = \eta$; $[b,\eta] = \rho(b)\eta$, $[K_b,\xi] = \rho(b)\xi$ with (R1), (R2).

Theorem 5.3 (Arity one, even generators, ch02 eq:c2-S1even, ch07 Def c7-vdot)**.** $S^{(1)}(v) = \int\big\{(v.A_m, A^*_m) + \frac{1}{4}(v.A_{ij}, A^{*ij}) - (v.\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (v.\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (v.c, c^*)\big\}$ with

$$v.A_m = v^n\partial_n A_m + A_n\partial_m v^n, \qquad v.A_{ij} = v^n\partial_n A_{ij} + \tfrac{1}{4}\operatorname{div} v\,A_{ij} - r^k{}_i A_{kj} - r^k{}_j A_{ik}, \qquad v.c = v^n\partial_n c,$$

$v.\lambda_\alpha{}^i = v^n\partial_n\lambda_\alpha{}^i - \frac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\lambda_\beta{}^i + \frac{3}{8}\operatorname{div} v\,\lambda_\alpha{}^i + r^i{}_j\lambda_\alpha{}^j$,
$v.\bar\lambda_{\dot\alpha i} = v^n\partial_n\bar\lambda_{\dot\alpha i} + \frac{1}{8}X_{mn}\bar\lambda_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} + \frac{3}{8}\operatorname{div} v\,\bar\lambda_{\dot\alpha i} - \bar\lambda_{\dot\alpha j}r^j{}_i$,

and $v.A^{ij} = v^n\partial_n A^{ij} + \frac{1}{4}\operatorname{div} v\,A^{ij} + r^i{}_k A^{kj} + r^j{}_k A^{ik}$. Dilatation (ch02 ex:c2-dil):

$$S^{(1)}(\Delta) = \int \big\{(2x\cdot\partial A_m + 2A_m, A^*_m) + \tfrac{1}{4}(2x\cdot\partial A_{ij} + 2A_{ij}, A^{*ij}) - (2x\cdot\partial\lambda_\alpha{}^i + 3\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (2x\cdot\partial\bar\lambda_{\dot\alpha i} + 3\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (2x\cdot\partial c, c^*)\big\}.$$

Theorem 5.4 (Arity one, odd generators, ch02 eq:c2-S1Q, eq:c2-S1S)**.**

$$S^{(1)}(\xi) = \int \Big\{-(\Gamma^m(\xi,\lambda), A^*_m) - \tfrac{1}{4}(\Gamma_{ij}(\xi,\lambda), A^{*ij}) + \tfrac{1}{2}\big((\rho(F_A)\xi)_\alpha{}^i, \lambda^{*\alpha}{}_i\big) - \tfrac{1}{2}\big((\rho(F_A)\xi)_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}\big)\Big\}, \quad \text{(F1)}$$

$$\Gamma^m(\xi,\lambda) = i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big), \qquad \Gamma_{kl}(\xi,\lambda) = 2\varepsilon_{jikl}\xi_\alpha{}^j\varepsilon^{\alpha\beta}\lambda_\beta{}^i - 2\big(\bar\xi_{\dot\alpha k}\varepsilon^{\dot\alpha\dot\beta}\bar\lambda_{\dot\beta l} - \bar\xi_{\dot\alpha l}\varepsilon^{\dot\alpha\dot\beta}\bar\lambda_{\dot\beta k}\big),$$

$$(\rho(F_A)\xi)_\alpha{}^i = -\tfrac{1}{2}F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i - \tfrac{1}{2}[A^{ik}, A_{kj}]\,\xi_\alpha{}^j - i\,\sigma^m_{\alpha\dot\alpha}\varepsilon^{\dot\alpha\dot\beta}D_m A^{ij}\,\bar\xi_{\dot\beta j},$$
$$(\rho(F_A)\xi)_{\dot\alpha i} = \tfrac{1}{2}F_{mn}\bar\xi_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} - \tfrac{1}{2}[A_{ik}, A^{kj}]\,\bar\xi_{\dot\alpha j} + i\,\varepsilon^{\alpha\beta}\sigma^m_{\alpha\dot\alpha}D_m A_{ij}\,\xi_\beta{}^j.$$

For $\eta\in\Sigma^*$, with $\epsilon = \rho(x)\eta$ of (K1) ($\xi = 0$),

$$S^{(1)}(\eta) = S^{(1)}(\xi)\big|_{\xi\to\epsilon(x)} + \int \Big\{\big(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \lambda^{*\alpha}{}_k\big) - \big(A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}, \bar\lambda^{*\dot\beta k}\big)\Big\}. \quad \text{(F2)}$$

Theorem 5.5 (Arity two, ch02 eq:c2-S2)**.** For ϵ_1, ϵ_2 each a constant ξ or $\rho(x)\eta$,

$$S^{(2)}(\epsilon_1,\epsilon_2) = \int \Big\{\tfrac{1}{4}\Big[-2i\,\Gamma^m(\epsilon_1,\epsilon_2)\big(\lambda^{*\alpha}{}_i, \sigma^m_{\alpha\dot\alpha}\bar\lambda^{*\dot\alpha i}\big) + \tfrac{1}{4}\Gamma_{kl}(\epsilon_1,\epsilon_2)\Gamma^{kl}(\lambda^*,\lambda^*)\Big] - \tfrac{1}{2}\big((\epsilon_1,\lambda^*),(\epsilon_2,\lambda^*)\big) - \big(\Gamma^m(\epsilon_1,\epsilon_2)A_m + \tfrac{1}{4}\Gamma_{ij}(\epsilon_1,\epsilon_2)A^{ij},\, c^*\big)\Big\}, \quad \text{(F3)}$$

$(\epsilon,\lambda^*) = \epsilon_\alpha{}^i\lambda^{*\alpha}{}_i - \bar\epsilon_{\dot\alpha i}\bar\lambda^{*\dot\alpha i}$, $\Gamma^{kl}(\lambda^*,\lambda^*) = -2\varepsilon^{klpj}\big(\varepsilon_{\alpha\beta}\lambda^{*\beta}{}_j, \lambda^{*\alpha}{}_p\big) + 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\lambda^{*\dot\alpha l}, \bar\lambda^{*\dot\beta k}\big) - 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\lambda^{*\dot\alpha k}, \bar\lambda^{*\dot\beta l}\big)$ ((G4) with $\mathfrak{g}$ contracted), Γ^m, Γ_{kl} of (G1), (G2).

Corollary 5.6 (Closure on fields)**.** With $\xi.A = -\Gamma(\xi,\lambda)$, $\xi.\lambda = -\frac{1}{2}\rho(F_A)\xi$ (the field coefficients in (F1), sign pattern of Theorem 5.3), abelian free fields:

$$\{\xi_a., \xi_b.\}A_m = -v^n F_{nm} + v^I\partial_m A_I, \qquad \{\xi_a., \xi_b.\}A_{ij} = -v^n\partial_n A_{ij}, \qquad v = \Gamma(\xi_a,\xi_b).$$

6 Step-1 results (ch04, ch05, ch06)

Unit: $\dfrac{\hbar\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y)}{16\pi^2(X,Y)}$ (ch04); $\int \equiv \int d^4x$.

Theorem 6.1 (Translated Step-1 statements)**.**

- Quadratic counterterm (ch04 thm:c4-quadratic):

$$\operatorname{unit}\Big\{\tfrac{9}{4}\varepsilon^{-1}\int\Big[(A_m,A_m) + \tfrac{1}{4}(A^{ij},A_{ij})\Big] + \log\varepsilon\Big[-\tfrac{1}{2}\int(F^0_{mn},F^0_{mn}) - \tfrac{1}{4}\int(\partial_m A^{ij},\partial_m A_{ij}) + 8i\int(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i) - \int(\partial_m c, A^*_m)\Big]\Big\},$$

  $F^0_{mn} = \partial_m A_n - \partial_n A_m$; coefficient of $\int(\partial_m A_m, \partial_n A_n)$ equal to 0.

- Logarithmic part (ch04 thm:c4-log): unit $\log\varepsilon\,(Q+\{I,-\})$ of the field-counting functional

$$2\int\Big[(A_m,A_m^*)+\tfrac{1}{4}(A^{ij},A_{ij}^*)\Big]-4\int\Big[(\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i)-(\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i})\Big]-\int(c^*,c).$$

- Mass-term variation (ch04 rem:c4-epsinv): $-\frac{9}{2}\,\mathrm{unit}\,\varepsilon^{-1}\int(\partial_m c,A_m)$. Obstruction and finite counterterm (ch04 eq:c4-obstruction, eq:c4-finite):

$$-\tfrac{27}{8}\,\mathrm{unit}\int(\partial^2 c,\partial_m A_m)=\tfrac{27}{16}\,\mathrm{unit}\,Q\int(\partial_m A_m,\partial_n A_n),\qquad I\mapsto I-\tfrac{27}{16}\,\mathrm{unit}\int(\partial_m A_m,\partial_n A_n).$$

- Scalar bilinears (ch05 def:c5-bilinear, thm:c5-Oeps): $\int\omega(\iota_{w_1}A,\iota_{w_2}A)=\frac{1}{16}\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})$, $(w_1,w_2)=\frac{1}{4}w_1^{ij}w_{2ij}$; trace (Konishi) $\int\omega(A|_{W^\perp},A|_{W^\perp})=\frac{1}{4}\int\omega(A^{ij},A_{ij})$; $\mathbf{20'}$: $\frac{1}{16}\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})-\frac{1}{96}w_1^{ij}w_{2ij}\int\omega(A^{kl},A_{kl})$;

$$\partial_\delta I^{CT}[\varepsilon]=\mathrm{unit}\,\log\varepsilon\Big[\tfrac{1}{4}\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})-\tfrac{1}{8}\,w_1^{ij}w_{2ij}\int\omega\,(A^{kl},A_{kl})\Big]+\frac{\hbar\dim\mathfrak{g}\,w_1^{ij}w_{2ij}}{32\pi^2\,\varepsilon}\int\omega.$$

- Dilatation coefficients (ch06 thm:c6-dil, cor:c6-dilcomp), $\Delta=2E$: $[(d_{CE}+Q)O[\infty]](\Delta)=2(E+w_O+\text{coefficient})O[\infty]+\dots$ with coefficient, in units: $\mathbf{20'}$: 0; Konishi $\frac{1}{4}\int\omega(A^{ij},A_{ij})$: -24; $\int\omega\,(X,A_{ij})$: -4; $\int\omega(X,A_m)$: -4; $\int\omega(\zeta,\lambda)=\int\omega\big[(\lambda_\alpha{}^i,\zeta^\alpha{}_i)-(\bar\lambda_{\dot\alpha i},\bar\zeta^{\dot\alpha i})\big]$: -8; $\int\omega(X,c)$: -2. For $\mathfrak{su}(N)$: unit $=-\frac{g^2N}{32\pi^2}$, Konishi $-24\cdot\mathrm{unit}=\frac{3g^2N}{4\pi^2}$ (ch06 rem:c6-konishi).

7 Step-2 bases (DESIGN_STEP2.md, Part C)

Definition 7.1 (Two-component bases)**.**

$$G_1=\int([c,A|_V],d^*dA|_V)=-\int([c,A_n],\partial_m F^0_{mn}),$$
$$G_2=\int([c,A|_V],dd^*A|_V)=-\int([c,A_n],\partial_n\partial_m A_m),$$
$$G_3=\int([c,A|_{W^\perp}],d^*dA|_{W^\perp})=-\tfrac{1}{4}\int([c,A^{ij}],\partial^2 A_{ij}),$$
$$G_4=\int([c,dc],A^*)=\int([c,\partial_m c],A_m^*),$$
$$G_5=\int([c,\lambda],\not{\partial}\lambda)=i\int\Big\{\big([c,\lambda_\alpha{}^i],\bar\sigma^{m\dot\alpha\alpha}\partial_m\bar\lambda_{\dot\alpha i}\big)+\big([c,\bar\lambda_{\dot\alpha i}],\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i\big)\Big\},$$

$$H_1=\int(dA|_V,[A|_V\wedge A|_V])=2\int(\partial_m A_n,[A_m,A_n]),$$
$$H_2=\int(\partial_m A_I,[A_m,A_I])=\tfrac{1}{4}\int(\partial_m A^{ij},[A_m,A_{ij}]),$$
$$H_3=\int([A|_V,c],A^*|_V)=\int([A_m,c],A_m^*),$$
$$H_4=\int([A|_{W^\perp},c],A^*|_{W^\perp})=\tfrac{1}{4}\int([A^{ij},c],A_{ij}^*),$$
$$H_5=\int([c,\lambda],\lambda^*)=\int([c,\lambda_\alpha{}^i],\lambda^{*\alpha}{}_i)-\int([c,\bar\lambda_{\dot\alpha i}],\bar\lambda^{*\dot\alpha i}),$$
$$H_6=\int([c,c],c^*).$$

H_2 is read componentwise, $\sum_{m,I}(\partial_m A_I,[A_m,A_I])$; the two-form reading $\int(dA|_{W^\perp},[A\wedge A]|_{V\times W^\perp})$ equals $2H_2$. Free counterterm directions: $\beta_1\int(dA|_V,dA|_V)=\frac{\beta_1}{2}\int(F^0_{mn},F^0_{mn})$, $\beta_2\int(dA|_{W^\perp},dA|_{W^\perp})=\frac{\beta_2}{4}\int(\partial_m A^{ij},\partial_m A_{ij})$, $\beta_3\int(\lambda,\not{\partial}\lambda)=2i\beta_3\int(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i)$, $\beta_4\int(\partial_m c,A_m^*)$; finite $\frac{27}{16}\,\mathrm{unit}\int(\partial_m A_m,\partial_n A_n)$.

8 Weinberg column

object	Srednicki form (this dictionary)	Weinberg form
Clifford action on the Dirac column $(\psi_\alpha, \bar\chi^{\dot\alpha})$	$\rho(v) = -i\gamma_S^\mu v_\mu\vert_{\text{Wick}}$	$\rho(v) = \gamma_W^\mu v_\mu\vert_{\text{Wick}}$, $\{\gamma_W^\mu, \gamma_W^\nu\} = 2\eta^{\mu\nu}$ (D1)
chirality of S_+	$\gamma_{5S} = -1$	$\gamma_{5W} = +1$, same projector P_L (D2)
spinor metric	$\varepsilon^{\alpha\beta}$, raise/lower	e_{ab} inserted, $e_{ab} = \varepsilon^{ab}$ numerically (D7)
two-component supercharge	Q_α	$\mathcal{Q}_a(W) = Q^{\dagger\dot a}(S)$, $\mathcal{Q}_a^*(W) = Q^a(S)$ (D7)
gauge field, coupling	$A_m = -igA_m^a T^a$, $D = \partial - igA^a T^a$	$t_\alpha = gT^a$, $A_W^\alpha = A^a$, $D = \partial - it_\alpha A^\alpha$ (D9)

9 Verification (P4)

`check_dictionary.py` (log `check_dictionary.log`, table `check_results.json`, entries with residuals `dictionary.json`) evaluates each displayed formula with spinor, $SU(4)$ or bispinor structure against the index-free formula in the realisation of engine A (the vector-index entries of Proposition 4.3 are the manuscript's formulas with $d = dx^m\partial_m$, $d^* = -\partial_m$ on one-forms, checked in T3, H-G1, H-G2; rows 3–5 of the Weinberg column are the database entries D7, D9), on random pointwise jets (pointwise identities) and plane waves with $\sum p = 0$ (integrated identities), Grassmann fields by polarisation ($\theta_1\theta_2$ coefficient $B(u_1, u_2) - B(u_2, u_1)$), $\mathfrak{g} = \mathfrak{su}(2)$ with $(X_a, X_b) = 0.83\,\delta_{ab}$.

checks	content
W1–W8	Srednicki $\bar\sigma = \varepsilon\sigma\varepsilon$; Wick map; Euclidean Clifford; Σ^I identities and Fierz; $F_{\mu\nu}F^{\mu\nu} = F_{mn}F_{mn}$
R1–R7	Clifford, symmetry of $(-,-)$, Assumption 3.1, chirality, $\rho = \gamma_W$ on the Dirac column
B0–B2, A0–A3	equality with engine B; intertwiner with engine A = classical model
S1–S5, G1–G8	bispinor scalars, reality, quartic identity; Γ (G1)–(G4) and suppressed forms; square-zero elements
L1–L7	action (pointwise and integrated, Grassmann), Yukawa suppressed form
Q1–Q4, P1–P3	Q, Q^{GF}, H; propagators in both realisations
E-P, E-M, E-D, E-K, E-R	$S^{(1)}(v)$ for all even generator types
O1–O4, K1–K4	$S^{(1)}(\xi)$, $S^{(1)}(\eta)$, $S^{(2)}$ (three cases), brackets (B1)–(B3)
T1–T10, H-G1–H-H5	Step-1 functionals, field-counting functional, bilinears; G_1–G_5, H_1–H_5
RT1, RT2	round trips: spinor and $(\lambda, \not{d}\lambda)$ bilinear form, index-free → two-component → index-free
C1–C6	real-slice negativity of the scalar kinetic term; closure (Cor. 5.6); [ESW, Prop 2.1]; canonical $\mathfrak{su}(N)$ map; Konishi number
N1–N6	negative controls: sign-flipped entries fail

Result: 118/118 passed, maximal relative residual 2.9×10^{-12}.